# Kepler’s laws

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use orbital shape, equal areas and period comparisons.

**SUCCESS CRITERIA - I can...**

1. Place the central body at a focus of an ellipse.
2. Compare speed near and far from the central body.
3. Use $T^2=a^3$ only with the specified units and central body.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “period” and “ellipse” to explain your answer. Sentence starters are optional.

**period**: time for one complete cycle or orbit

**ellipse**: an oval-shaped closed curve with two focal points

**orbit**: a path around another body under gravity

## 01 Warm-up: bring back an earlier idea

Launch at **30 m/s** from level ground at **25°** or **65°**.
1. Use g=**10** and no drag.
2. **Compare ranges** and flight times.

**Sentence starter:** Both ... . They differ in ... because ... .

**Marzano check - circle:** ① ② ③

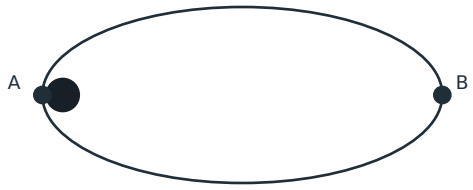
## 02 Worked example: explain the prediction

**Reference:** $T^2 = a^3$.
For Sun orbits only when T is in years and a is in AU. a is the semi-major axis. Equal areas are swept in equal times.

**Worked reasoning:** For an ideal Sun orbit, a=**4 AU** gives $T=\sqrt{(\mathbf{4^3})}=\mathbf{8}$ **years**. The orbit period depends on semimajor axis, not on a chosen point's instantaneous speed alone.

**Your turn:** Draw an ellipse, put the Sun at a focus and **mark the faster** and slower parts.

**Sentence starter:** In my work, I used ... to show ... . I labeled it ... .

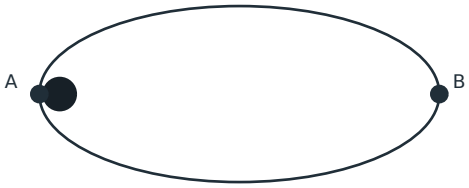



**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

**Mark the Sun** at a focus on an ellipse. **Label where the object** is fastest and slowest.

**Sentence starter:** In my work, I used ... to show ... . I labeled it ... .




**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A Sun-orbit has a=**1.42 AU**. Compute period. Then **find period** at a=**4 AU** and **explain why doubling a** does not double T.

**Sentence starter:** For this answer, I treated ... as true. I found ... . This supports ... because ... .

| Semimajor axis (AU) | Period (years) |
|---|---|
| 1.42 | ____ |
| 4 | ____ |

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Explain your reasoning

Does knowing an orbit’s period alone tell the date when an asteroid reaches Earth’s location? **What else** is needed?

**Sentence starter:** I notice … . This supports … because … .

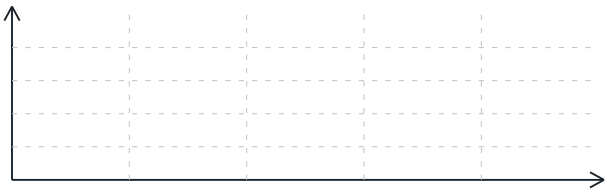
**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A ball begins with downward velocity **4 m/s** and falls **3 s**. Use downward positive, g=**10 m/s²** and no drag. **Find final velocity** and displacement.
**Sketch the motion** and **label a positive direction**.

**Sentence starter:** I found … [unit]. This value means … in this situation.

Velocity (m/s)



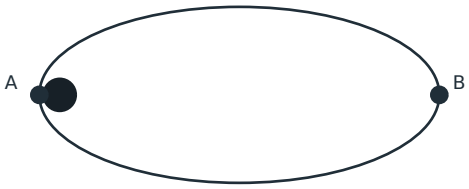
Time (s)

**Marzano check - circle:** 1 2 3

## 07 Exit check

For two Sun-orbits, $a_B$=**2** $a_A$.
1. Find $T_B/T_A$.
2. **State why the same unit** form cannot automatically be used for a Moon orbit about Earth.

**Sentence starter:** For this answer, I treated … as true. I found … . This supports … because … .



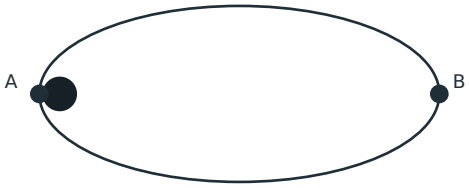

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

An ideal Sun orbit has period **27 years**. **Find its semimajor axis**. If eccentricity is **0.5**, find perihelion and aphelion distances using r=a(**1**∓e).

**Word help:** ideal = using stated simplifying conditions

**Sentence starter:** I found … [unit]. This value means … in this situation.




**Marzano check - circle:** 1 2 3